

Problem 1 : Basic Exam, Spring 2016

For $a < b$ real numbers and $f : [a, b] \rightarrow \mathbb{R}$ a function, do as follows:

- (a) Define what it means for f to be Riemann integrable on $[a, b]$.
(b) Let $\{x_n\}_{n=1}^{\infty} \subseteq [a, b]$ be a sequence such that $\lim_{n \rightarrow \infty} x_n$ exists and suppose that $f : [a, b] \rightarrow \mathbb{R}$ is defined by

$$f(x) = \begin{cases} 1, & \text{if } x \notin \{x_n\}_{n=1}^{\infty}, \\ 0, & \text{else.} \end{cases}$$

Using your definition, prove that f is Riemann integrable on $[a, b]$.

Solution. (a): $P \subset [a, b]$ is a **partition of $[a, b]$** if $P = \{x_1, \dots, x_n\}$ where $a = x_1 \leq x_2 \leq \dots \leq x_n = b$. Let

$$U(f, P) = \sum_{i=1}^{n-1} (x_{i+1} - x_i) \left(\sup_{x \in [x_i, x_{i+1}]} f(x) \right)$$

$$L(f, P) = \sum_{i=1}^{n-1} (x_{i+1} - x_i) \left(\inf_{x \in [x_i, x_{i+1}]} f(x) \right)$$

$$L(f) := \sup_{P \text{ partition}} L(f, P)$$

$$U(f) := \inf_{P \text{ partition}} U(f, P)$$

f is Riemann integrable if and only if $\forall \varepsilon > 0$ there exists a P such that

$$|L(f, P) - U(f, P)| < \varepsilon$$

(b): Let $z = \lim_{n \rightarrow \infty} x_n$. $\forall \varepsilon > 0$. There exists a $N \in \mathbb{N}$ such that $n \geq N \Rightarrow |x_n - z| < \frac{\varepsilon}{2}$. Let $I = [z - \frac{\varepsilon}{2}, z + \frac{\varepsilon}{2}] = [s, t]$. Let $I_k(\nu) = [x_k - \nu, x_k + \nu]$. Without loss of generality for $x_m \neq x_n$ for $m, n \in \{1, \dots, N\}$ and $x_m \in I$. Let $\nu \in (0, \frac{\varepsilon}{2(N)})$ small enough such that $I_m(\nu) \cap I_n(\nu) = \emptyset$ and $I_n(\nu) \cap I = \emptyset$. Again without loss of generality we will order the x_i and $x < x_i$. Note the following:

$$\sup_{x \in I_k(\nu)} f(x) = 1 = \sup_{x \in I} f(x)$$

$$\inf_{x \in I_k(\nu)} f(x) = 0 = \inf_{x \in I} f(x)$$

Thus only intervals which matter are $I, I_k(\nu)$

$$\begin{aligned} |L(f, P) - U(f, P)| &\leq |\varepsilon + 2N\nu| \\ &= \varepsilon + \varepsilon = 2\varepsilon \end{aligned}$$

Problem 2 : Basic Exam, Spring 2025

Let $\{f_n\}$ and $\{g_n\}$ be sequences of Riemann integrable functions $[0, 1] \rightarrow \mathbb{R}$ such that

1. There exists $C > 0$ such that

$$\int_0^1 |g_n(x)| \, dx \leq C, \quad n = 1, 2, \dots$$

2. f_n converges uniformly to f on $[0, 1]$.
3. For all $n \in \mathbb{N}$, $\lim_{m \rightarrow \infty} \int_0^1 |f_n(x)g_m(x)| \, dx = 0$.

Show that

$$\lim_{m \rightarrow \infty} \int_0^1 f(x)g_m(x) \, dx = 0.$$

Solution. $\forall \varepsilon > 0$:

By uniform convergence of $f_n \rightarrow f$, we have that $\exists N$ such that for all $n \geq N$ we have that $\|f - f_n\|_\infty < \frac{\varepsilon}{2}C$

$$\begin{aligned} \left| \int_0^1 f(x)g_m(x) \, dx - \int_0^1 f_n(x)g_m(x) \, dx \right| &= \left| \int_0^1 (f(x) - f_n(x))g_m(x) \, dx \right| \\ &\leq \int_0^1 |f(x) - f_n(x)||g_m(x)| \, dx \\ &\leq \|f - f_n\|_\infty \int_0^1 |g_m(x)| \, dx \\ &\leq C\|f - f_n\|_\infty < \frac{\varepsilon}{2} \end{aligned}$$

By assumption 3, there $\exists M \in \mathbb{N}$ such that $\forall m \geq M$ we have that

$$\begin{aligned} \int_0^1 |f_n(x)g_m(x)| \, dx &< \frac{\varepsilon}{2} \\ \left| \int_0^1 f(x)g_m(x) \, dx \right| &< \frac{\varepsilon}{2} + \left| \int_0^1 f_n(x)g_m(x) \, dx \right| \\ &\leq \frac{\varepsilon}{2} + \int_0^1 |f_n(x)g_m(x)| \, dx \\ &< \varepsilon. \end{aligned}$$

Thus

$$\lim_{n \rightarrow \infty} \int_0^1 f(x)g_m(x) \, dx = 0$$

Problem 3 : Basic Exam, Spring 2024

Let $f : [0, 1] \rightarrow (0, \infty)$ be a continuous function and let

$$M = \sup_{x \in [0, 1]} f(x).$$

Show that

$$\lim_{n \rightarrow \infty} \left(\int_0^1 (f(x))^n dx \right)^{1/n} = M.$$

Solution.

$$\begin{aligned} \left| \frac{1}{n} \log \left(\int_0^1 (f(x))^n dx \right) - \log(M) \right| &= \left| \frac{1}{n} \log \left(\frac{1}{M^n} \int_0^1 f(x)^n dx \right) \right| \\ &= \frac{1}{n} \left| \log \left(\frac{1}{M^n} \int_0^1 (f(x))^n dx \right) \right| \\ &\leq \\ &\int_0^1 (f(x))^n dx \leq M^n \end{aligned}$$

Thus the sequence

$$x_n = \left(\int_0^1 (f(x))^n dx \right)^{\frac{1}{n}} \leq M \Rightarrow \limsup_{n \rightarrow \infty} x_n \leq M$$

By compactness there exists a $x \in [0, 1]$ such that $f(x) = M$. $\forall \varepsilon > 0$, by continuity $\exists \delta > 0$ such that $\forall y \in (x - \delta, x + \delta)$ where $f(y) \geq M - \varepsilon$.

$$\begin{aligned} \left(\int_0^1 (f(x))^n dx \right)^{\frac{1}{n}} &= \left(\int_0^{x-\delta} (f(x))^n dx + \int_{x-\delta}^{x+\delta} (f(x))^n dx + \int_{x+\delta}^1 (f(x))^n dx \right)^{\frac{1}{n}} \\ &\geq \left(\int_0^{x-\delta} (f(x))^n dx + (2\delta)(M - \varepsilon)^n + \int_{x+\delta}^1 (f(x))^n dx \right)^{\frac{1}{n}} \\ &\geq (2\delta)^{\frac{1}{n}} (M - \varepsilon) \end{aligned}$$

Note that

$$\liminf_{n \rightarrow \infty} (2\delta)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} (2\delta)^{\frac{1}{n}} = 1.$$

Thus

$$\begin{aligned}\liminf_{n \rightarrow \infty} \left(\int_0^1 (f(x))^n \, dx \right)^{\frac{1}{n}} &\geq \liminf_{n \rightarrow \infty} ((2\delta)^{\frac{1}{n}} (M - \varepsilon)) \\ &= (M - \varepsilon)\end{aligned}$$

Since this is true for all $\varepsilon > 0$, we can take the limit as $\varepsilon \rightarrow 0^+$ which tells us that

$$\liminf_{n \rightarrow \infty} \left(\int_0^1 (f(x))^n \, dx \right)^{\frac{1}{n}} \geq M$$

Thus

$$M \leq \liminf_{n \rightarrow \infty} x_n \leq \liminf_{n \rightarrow \infty} x_n \leq M \Rightarrow \lim_{n \rightarrow \infty} \left(\int_0^1 (f(x))^n \, dx \right)^{\frac{1}{n}}$$